

UTKARSH BARNWAL

Greater Noida, Uttar Pradesh

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SUMMARY

Computer Science undergraduate focused on backend engineering and AI-powered systems. Experienced in building scalable microservices architectures, event-driven systems using RabbitMQ, and REST APIs with Node.js. Skilled in developing RAG-based AI applications and distributed backend systems.

EDUCATION

Lloyd Institute of Engineering and Technology

Bachelor of Technology (B.Tech) CGPA: 7.8

Jun 2022 – Jun 2026

Greater Noida, Uttar Pradesh

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Architecture: Microservices, Event-Driven Systems, RabbitMQ

Databases: MongoDB, Redis, SQL, Pinecone (Vector Database)

DevOps & Tools: Docker, Nginx, CI/CD, Git, GitHub, AWS (Basics)

AI & Systems: RAG, LLM APIs (OpenAI, Gemini), Custom AI Agents

PROJECTS

Aurora – LLM Powered RAG System | MERN, Redis, RAG, Gemini API, Pinecone

June 2025

- Engineered a production-ready RAG-based AI assistant enabling context-aware responses over custom datasets.
- Implemented semantic vector search using Pinecone to enhance retrieval precision and reduce hallucinated outputs.
- Designed modular REST APIs handling AI inference, query processing, and retrieval workflows.
- Implemented JWT-based authentication and role-based access control for secure multi-user access.

E-Commerce Backend with AI Agent | Node.js, Express, MongoDB, Redis, Docker

July 2025

- Architected a distributed microservices backend managing product, cart, order, and authentication services.
- Developed 20+ REST APIs with optimized MongoDB schemas ensuring scalability and data consistency.
- Implemented event-driven communication using RabbitMQ for asynchronous inter-service messaging and notification processing.
- Built an AI automation agent simulating transactional user workflows and session orchestration.
- Containerized services using Docker to ensure environment consistency and deployment portability.

CERTIFICATIONS

- Backend Development (Node.js, Express.js, MongoDB) – Sheryians
- Projects & Hackathons – Sheryians